

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Capstone Project — Final Report

Intern Name	Upputholla Gopi
Institution	VVIT, Guntur
Domain	Mutual Fund / Fintech
Duration	7 Working Days June 2026
Technologies	Python, SQL, Power BI

1. Executive Summary

This capstone project presents a full-stack Mutual Fund Analytics Platform developed for Bluestock Fintech. The platform ingests publicly available data from AMFI India and mfapi.in, processes it through a robust ETL pipeline, stores it in a normalized SQLite database, and delivers insights through an interactive Power BI dashboard.

Key Achievements:

- Built automated ETL pipeline processing 87,000+ rows of financial data
- Designed 5-table star schema SQLite database with 8 fact/dimension tables
- Created 10+ EDA charts revealing key market trends and investor behaviour
- Computed Sharpe, Sortino, Alpha, Beta, VaR metrics for 40 fund schemes
- Built 4-page interactive Power BI dashboard with Bluestock branding
- Developed fund recommendation system based on risk appetite

2. Data Sources & Datasets

All data is sourced from publicly available APIs and reports published by AMFI India, NSE, and BSE. No proprietary data was used.

Dataset	Rows	Description
01_fund_master.csv	40	Master list of 40 fund schemes
02_nav_history.csv	46,000+	Daily NAV 2022-2026
03_aum_by_fund_house.csv	90	Quarterly AUM by AMC
04_monthly_sip_inflows.csv	48	Monthly SIP inflow data
05_category_inflows.csv	144	Net inflows by category
06_industry_folio_count.csv	21	Total MF folios growth
07_scheme_performance.csv	40	Risk-return metrics
08_investor_transactions.csv	32,000+	SIP/Lumpsum transactions
09_portfolio_holdings.csv	322	Equity fund holdings
10_benchmark_indices.csv	8,050	Nifty/BSE index data

3. EDA Key Findings

NAV Trend: All 40 funds showed consistent growth 2022-2026 with strong 2023 bull run.

SBI Dominance: SBI Mutual Fund leads AUM at Rs.12.5 Lakh Crore — nearly 2x ICICI Pru.

SIP All-time High: Monthly SIP inflows hit Rs.31,002 Crore in Dec 2025.

Age Demographics: 26-35 age group forms the largest investor segment.

Geographic: Maharashtra and Karnataka lead in total investment amounts.

T30 vs B30: T30 cities contribute ~70% of total investments.

Folio Growth: Total folios doubled from 13.26 Cr to 26.12 Cr — 97% growth.

Correlation: Large cap funds show 0.85+ correlation with each other.

Sector: Consumer Goods and IT together account for ~25% of portfolios.

Gender Gap: Male investors outnumber female 65:35.

4. Performance Analytics

Key risk-return metrics were computed for all 40 fund schemes using daily NAV history from January 2022 to May 2026.

Metric	Formula	Purpose
CAGR	$(NAV_end/NAV_start)^{(1/n)} - 1$	Annualised return
Sharpe Ratio	$(R_p - R_f) / Std(R_p) \times \sqrt{252}$	Risk-adjusted return
Sortino Ratio	$(R_p - R_f) / Downside_Std$	Downside risk measure
Alpha	OLS intercept x 252	Return above benchmark
Beta	OLS slope	Market sensitivity
Max Drawdown	$\min(NAV/\max_NAV - 1)$	Worst peak-to-trough
VaR (95%)	5th percentile of returns	Daily loss threshold
CVaR	Mean of returns < VaR	Expected tail loss

5. Recommendations

- Investors with Low risk appetite should consider Liquid and Short-term Debt funds with Sharpe > 1 .
- Moderate risk investors can explore Large Cap funds with consistent 3yr CAGR $> 12\%$.
- High risk investors may consider Mid/Small Cap funds but should monitor Max Drawdown.
- AMCs should target B30 cities for SIP campaigns as growth potential is higher.
- Funds with high HHI concentration should diversify sector exposure.
- At-risk SIP investors (gap > 35 days) need re-engagement through notifications.

6. Limitations

- Investor transaction data is synthetically generated for educational purposes.
- NAV data anchored to real values but simulated forward with realistic parameters.
- Analysis limited to 40 schemes out of 1,908 total industry schemes.
- Dashboard requires Power BI Desktop to view interactively.